

AtlasMSI

Autoencoder-Based Representation Learning and Segmentation for MALDI-MSI

Max Stróżyk

2026-09-25

Welcome

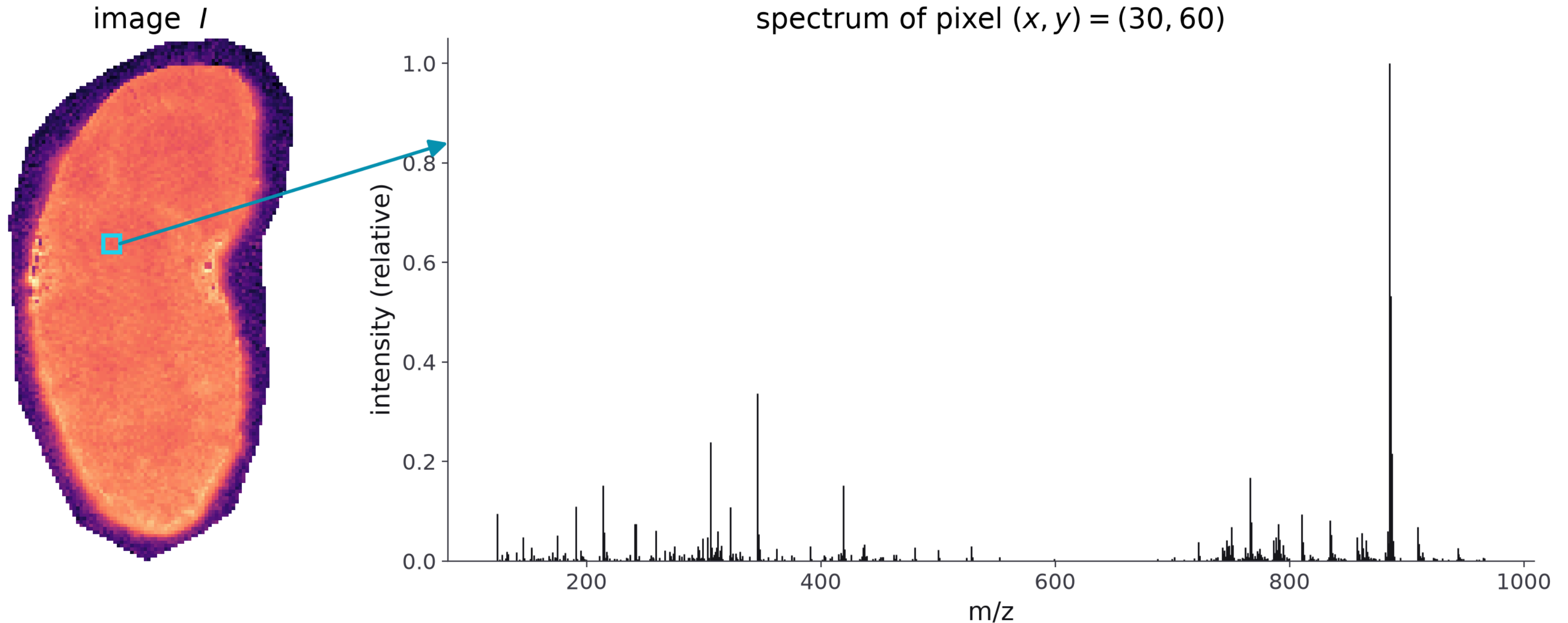
Welcome, BioGeMT attendees

Special thanks to Michał Ciach for the invitation.

Introduction

Data: mass spectrometry images (MALDI)

INTUITION



- every pixel of the image carries a whole mass spectrum
- MALDI: the tissue section is scanned pixel by pixel; one spectrum per position

Data: mathematical description

FORMAL

Image $I \subset \mathbb{N}^2$ (or $\mathbb{N}^3$), pixel $p = (x, y)$

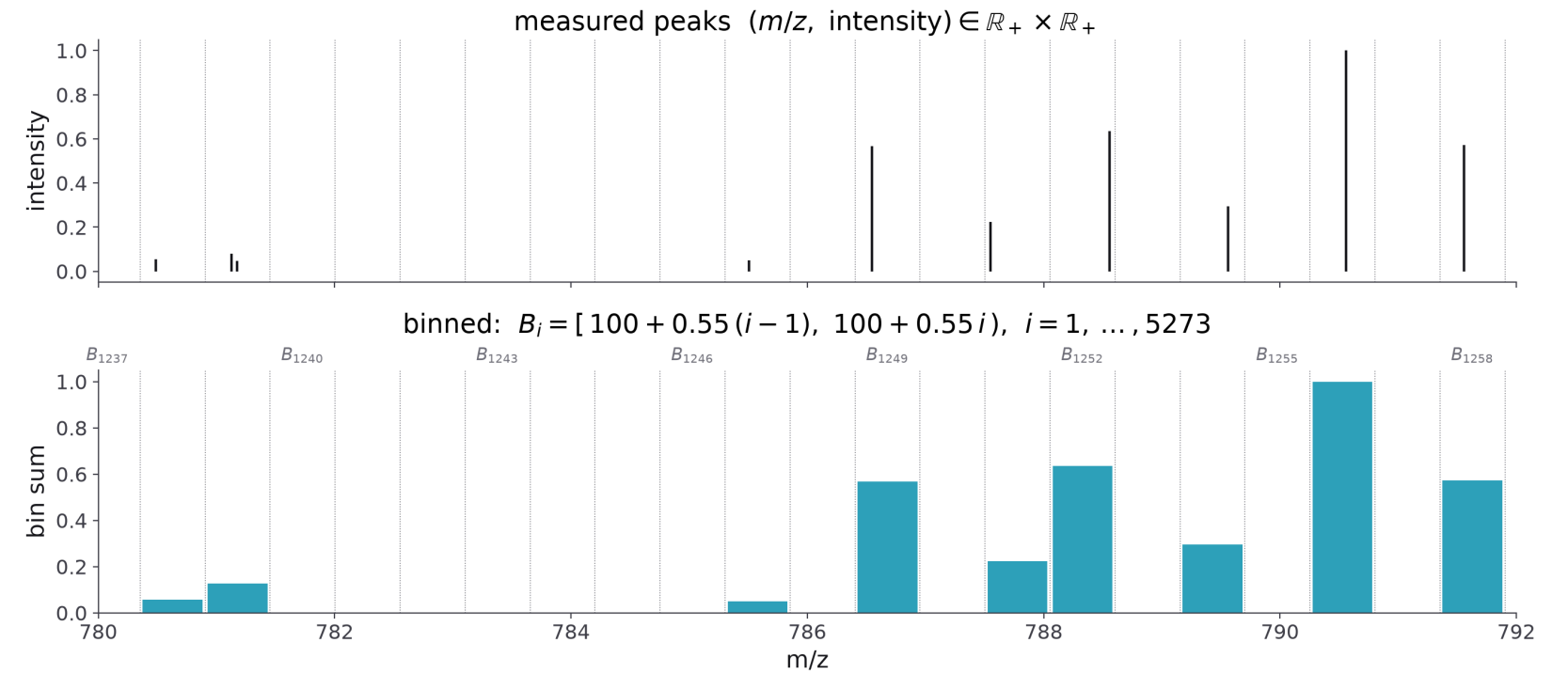
Spectrum at p — peaks $(m, v) \in \mathbb{R}_+ \times \mathbb{R}_+$

$$\mu_p = \sum_k v_k \delta_{m_k}$$

Binning of the m/z axis $[100, 3000]$ with step 0.55

$$B_i = [100 + 0.55(i - 1), 100 + 0.55i), \quad i = 1, \dots, M$$

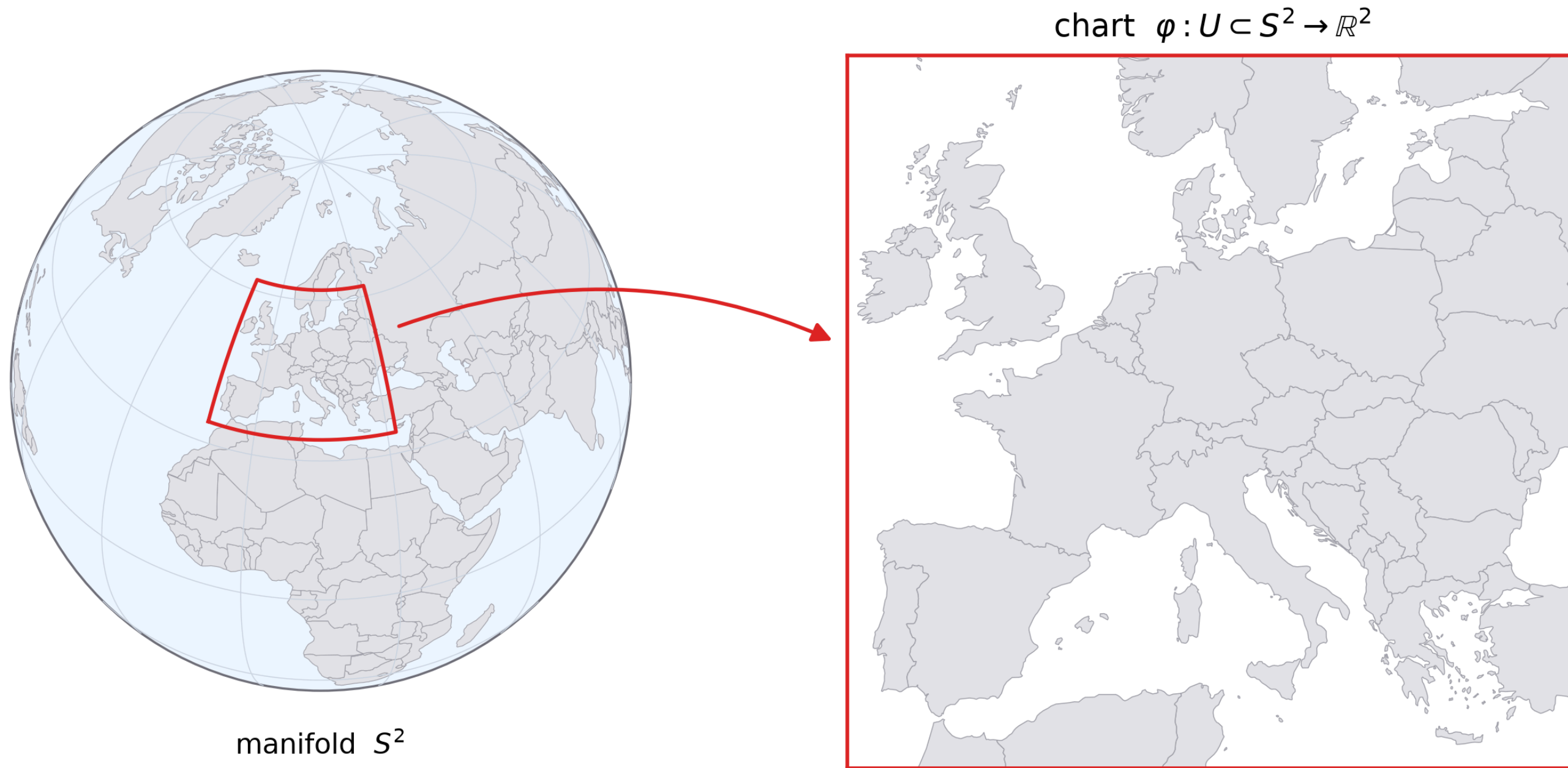
$$x_{p,i} = \sum_{m_k \in B_i} v_k, \quad M = 5273$$



$$\text{data} \subset I \times \prod_{i=1}^M (B_i \times \mathbb{R}_+) \cong I \times \mathbb{R}_+^M, \quad x_p \mapsto \frac{x_p}{\|x_p\|_1} \in \Delta^{M-1} \quad (\text{TIC})$$

Segmentation: where the word *atlas* comes from

INTUITION



- the globe is known, but we work with **charts**: approximate, direct representations in $\mathbb{R}^2$
- an **atlas** is a collection of charts $(U_\alpha, \varphi_\alpha)$, $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^d$

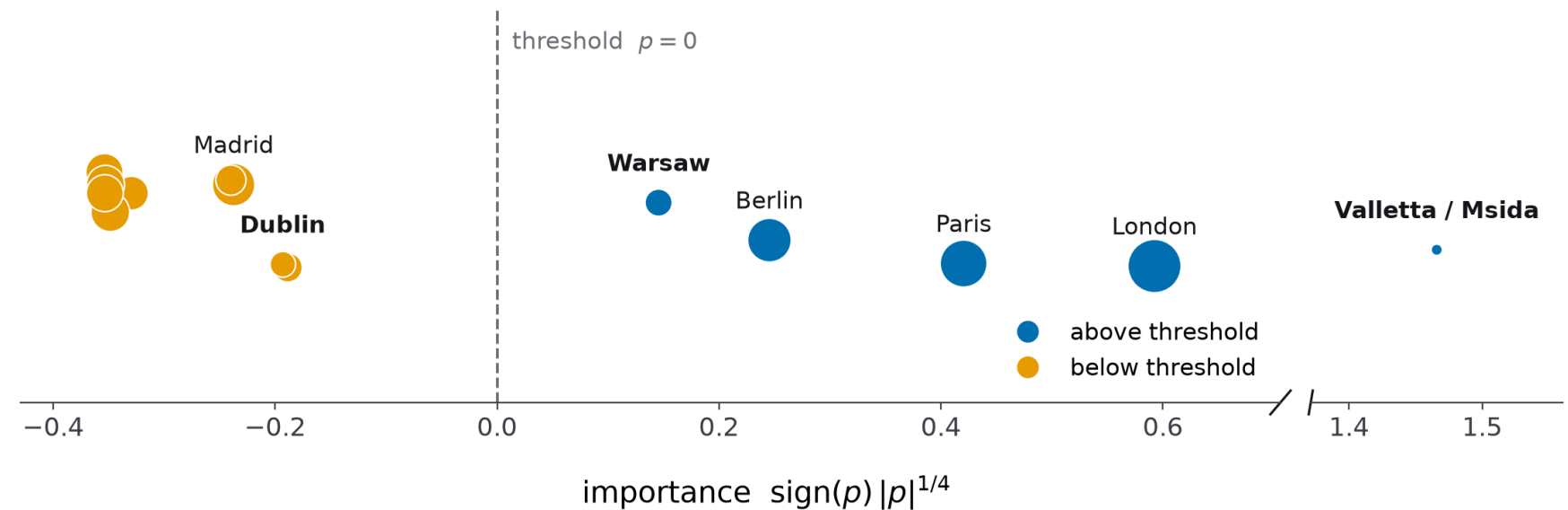
Segmentation: a trivial practical example

capitals: position $(\lambda, \phi) \in \mathbb{R}^2$, marker area = publications



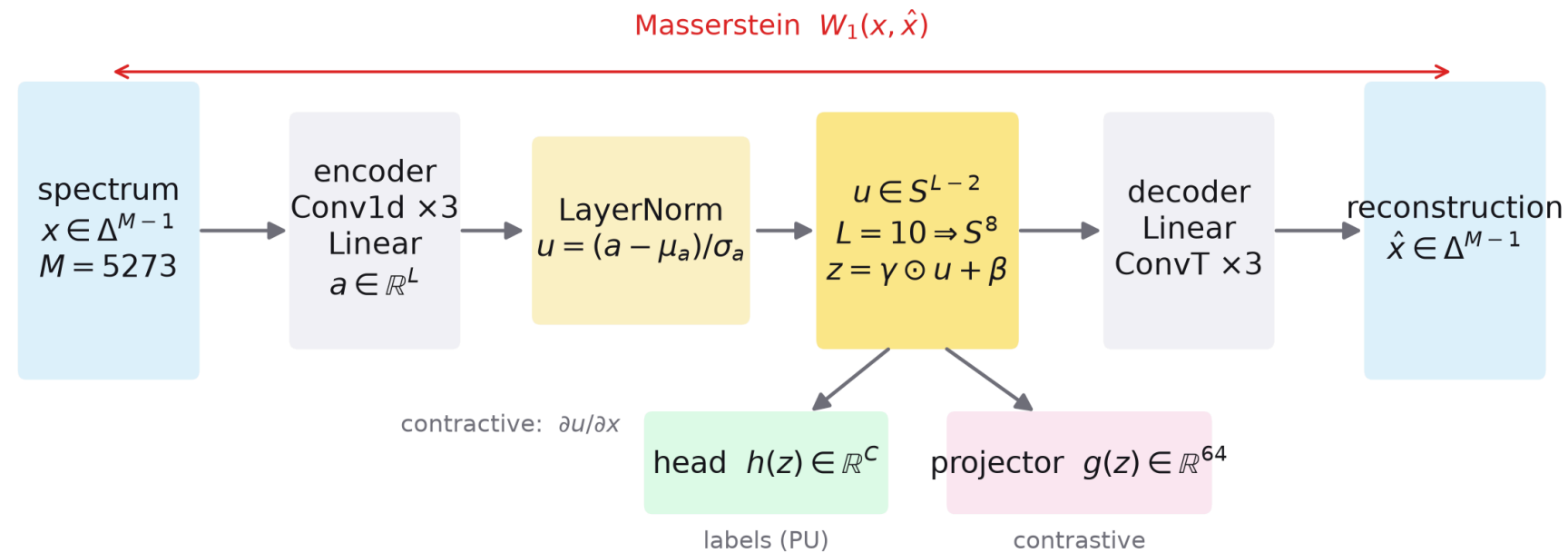
- a capital needs several coordinates (position, name, country, ...)
- we ask **one** question: how active is mass spectrometry there?
- PubMed publications with a capital-city affiliation, normalized over the shown capitals

$$p(x) = (x - r_1)(x - r_2)(x - r_3)(x - r_4), \quad x = \log_{10} n, \quad n = \text{publications}$$

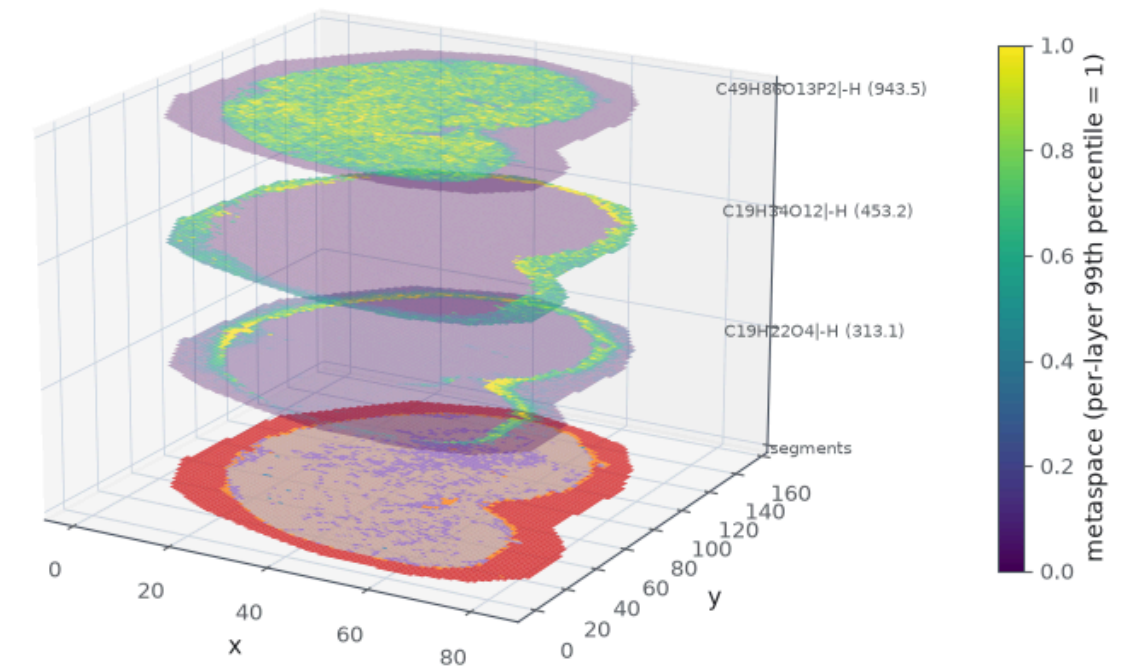


- representation reduced to $\mathbb{R}$ — two groups appear

Segmentation: the analysed example



metaspace_spearman: 2024-02-20_01h46m58s, layers = metaspace



- encoder compresses a spectrum into a latent point z
- decoder reconstructs the spectrum from z
- goal: construct z so that it is **stable** and **meaningful**

Segmentation: the atlas, formally

FORMAL

$$\text{Enc} : \Delta^{M-1} \rightarrow S^{L-2}(\sqrt{L}) \subset \mathbf{1}^\perp \subset \mathbb{R}^L, \quad \text{Dec} : \mathbb{R}^L \rightarrow \Delta^{M-1}$$

Bottleneck LayerNorm of $a = \text{CNN}(x) \in \mathbb{R}^L$

$$u = \frac{a - \mu_a \mathbf{1}}{\sigma_a} \Rightarrow \mathbf{1}^\top u = 0, \quad \|u\|^2 = L$$

so u lies on the sphere $S^{L-2}(\sqrt{L})$ in $\mathbf{1}^\perp$; $z = \gamma \odot u + \beta$ is an affine image of it.

$$L = 10 \Rightarrow u \in S^8$$

Heads on the latent

- decoder $\hat{x} = \text{Dec}(z)$
- predictive head $h(z) \in \mathbb{R}^C$ (annotated ions)
- projector $g(z) \in \mathbb{R}^{64}$ (contrastive)

Data inside the sphere: neighbourhood-based (TwoNN)
intrinsic dimension of the latent points ≈ 5.1 -5.6

Components

The example we will develop

TOY MODEL

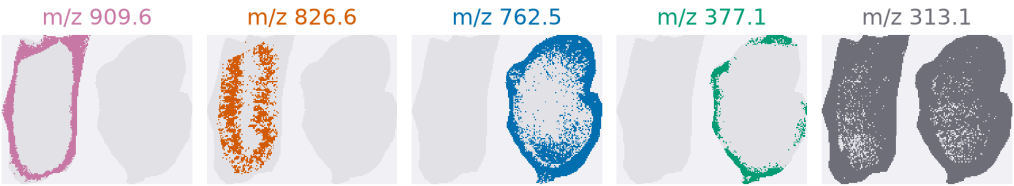
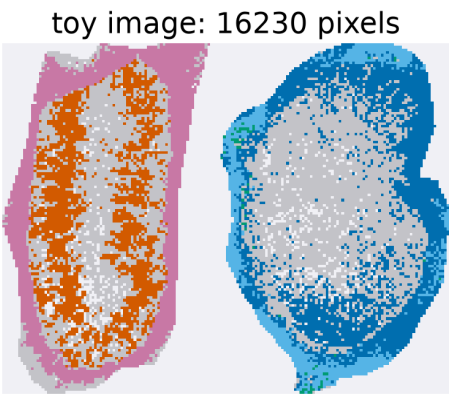
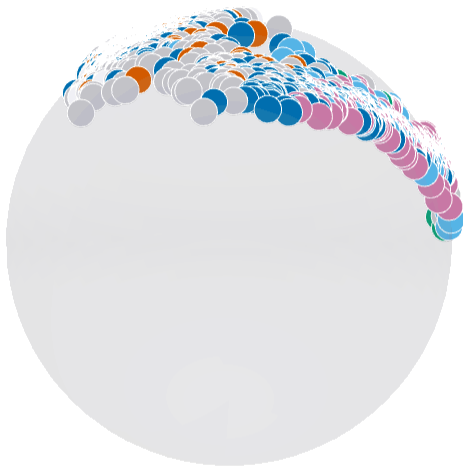
- two kidney sections of one series (METASPACE d28-2017-2, d28-2006-2): **16 230 pixels**
- m/z 200-950, step 0.55: **1364 bins; 21 annotated ions** (7 shared)
- same architecture and optimizer as the real model, **10 epochs**
- latent $L = 3 \Rightarrow u \in S^1$ (a circle), $L = 4 \Rightarrow u \in S^2$ (a sphere)
- components: **Masserstein, predictive head, contrastive, contractive**
- loss weights as in the real model: head 0.2, contrastive 0.001, contractive 0.002

The toy latent, interactively



reconstruction · sample ▼

- 909.6
- 826.6
- 762.5 + 377.1
- 762.5
- 377.1
- 313.1 only
- none of the five

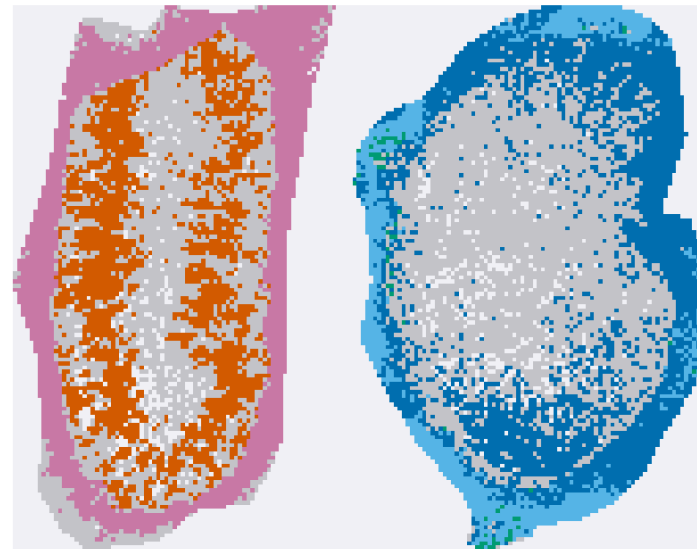
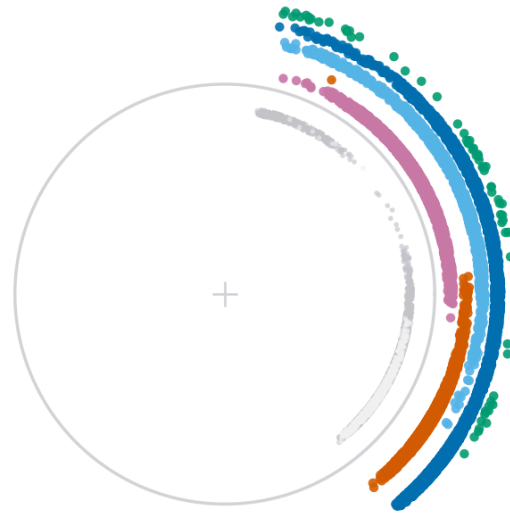


909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

Region key (left): 909.6 · 826.6 · 762.5 + 377.1 · 762.5 · 377.1 · 313.1 only · none of the five. **Ion maps (right):** panel title = m/z; ion-coloured pixels are annotated, grey pixels are not.

The toy latent: reconstruction only — $\mathcal{S}^1$

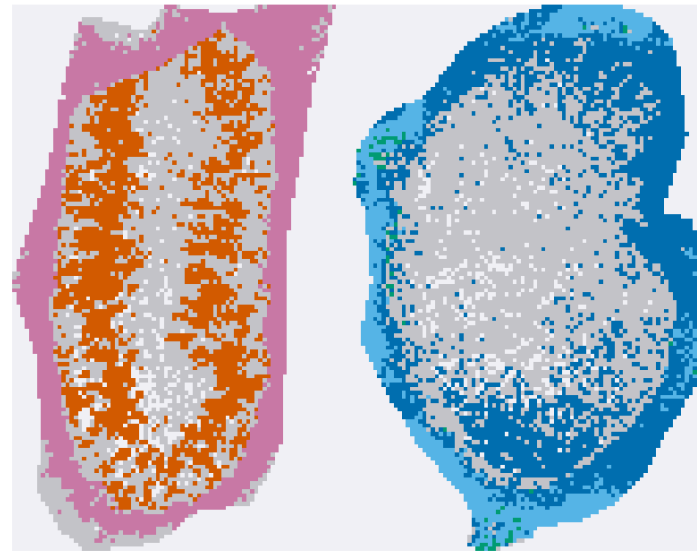
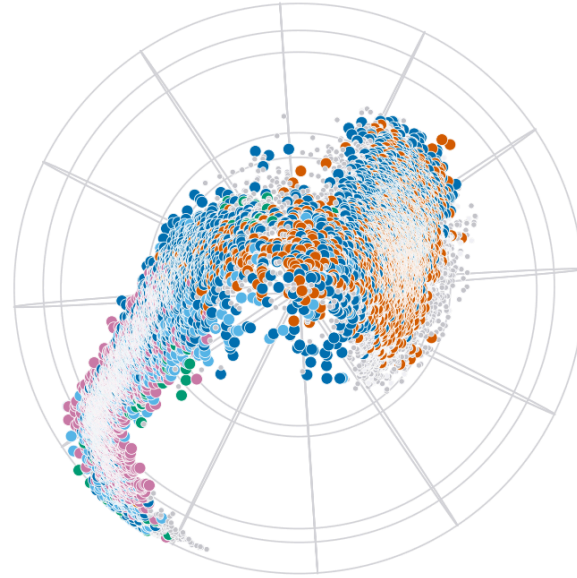
reconstruction



909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

The toy latent: reconstruction only — $\mathcal{S}^2$

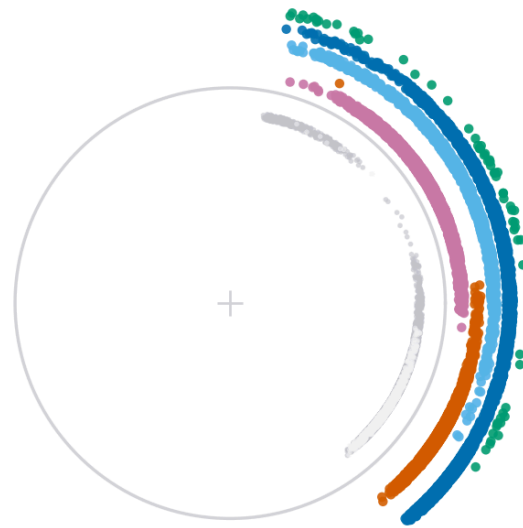
reconstruction



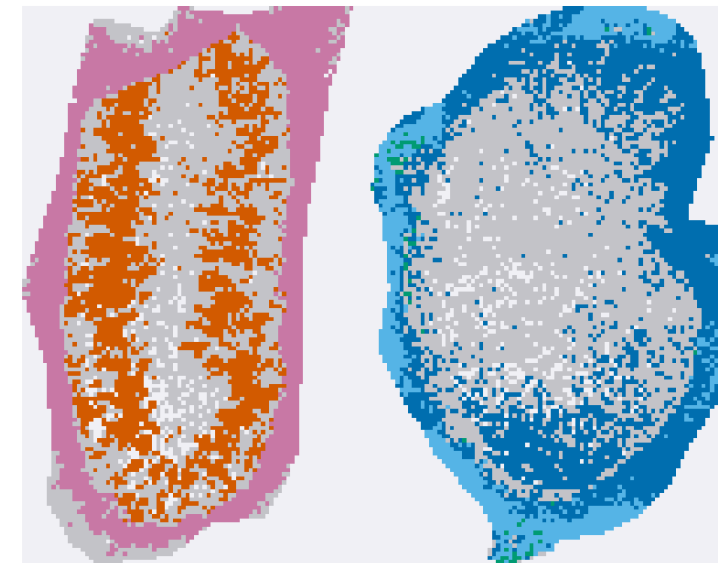
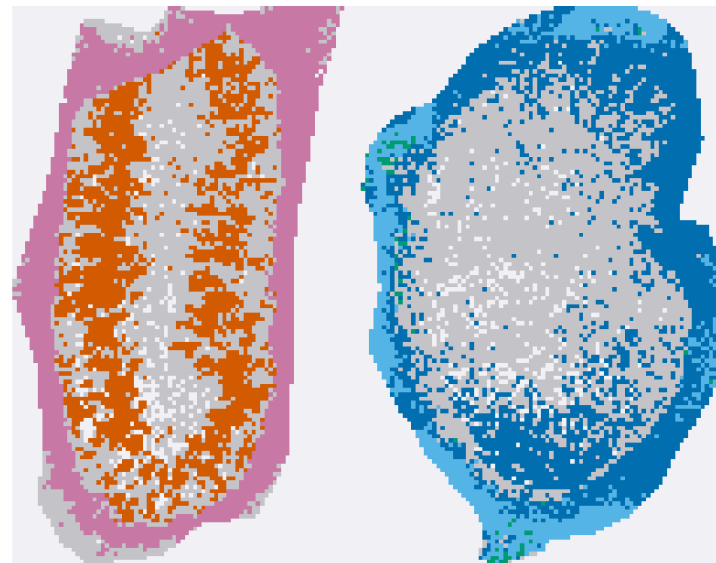
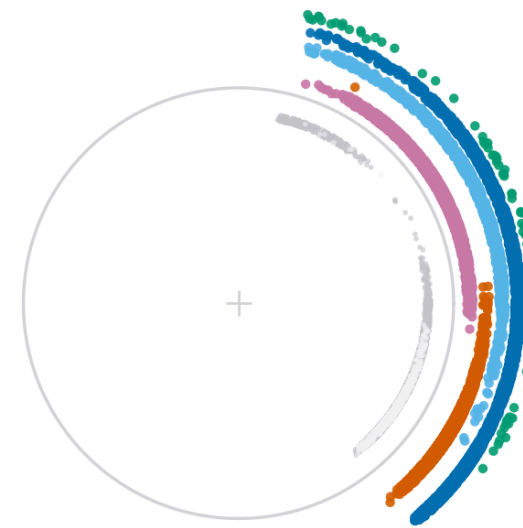
909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

Predictive head — $\mathcal{S}^1$

reconstruction



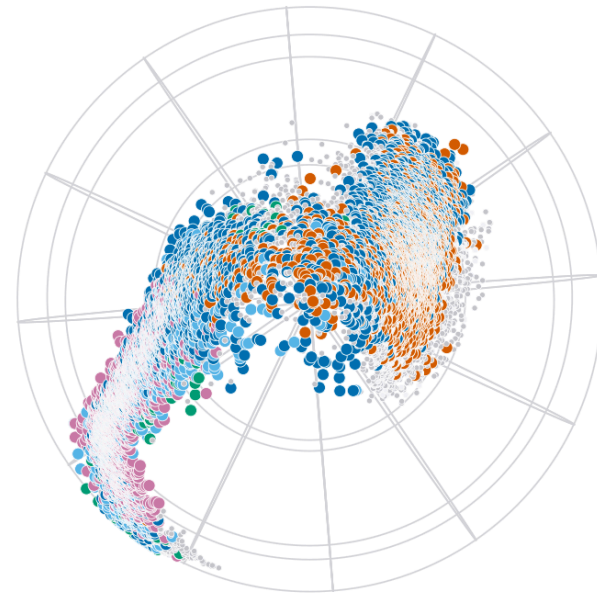
+ head



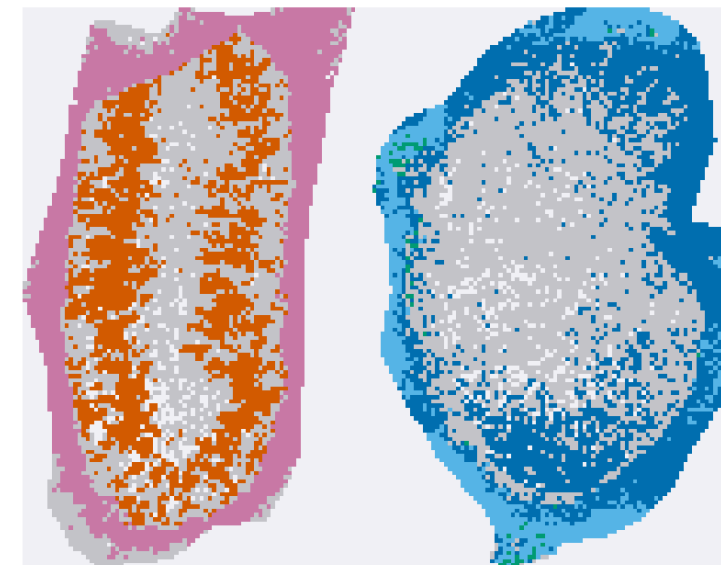
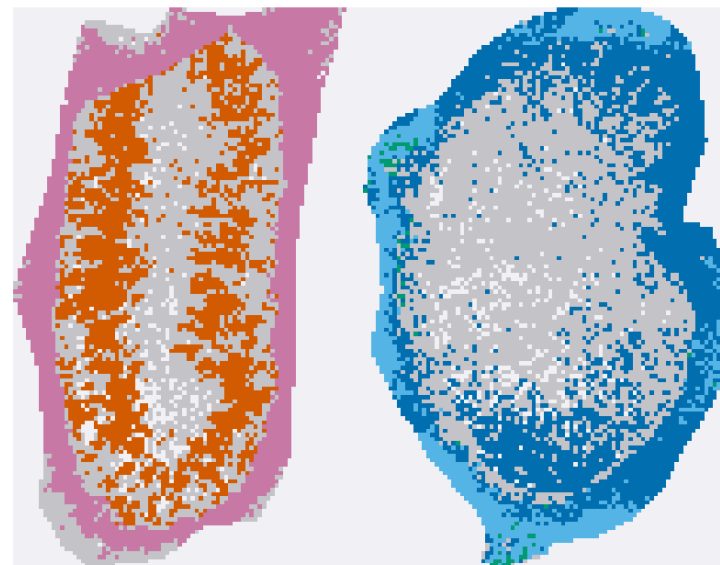
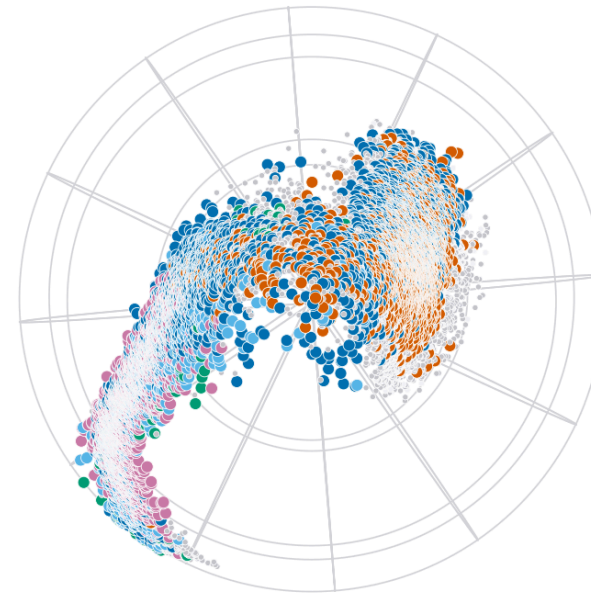
909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

Predictive head — $\mathcal{S}^2$

reconstruction

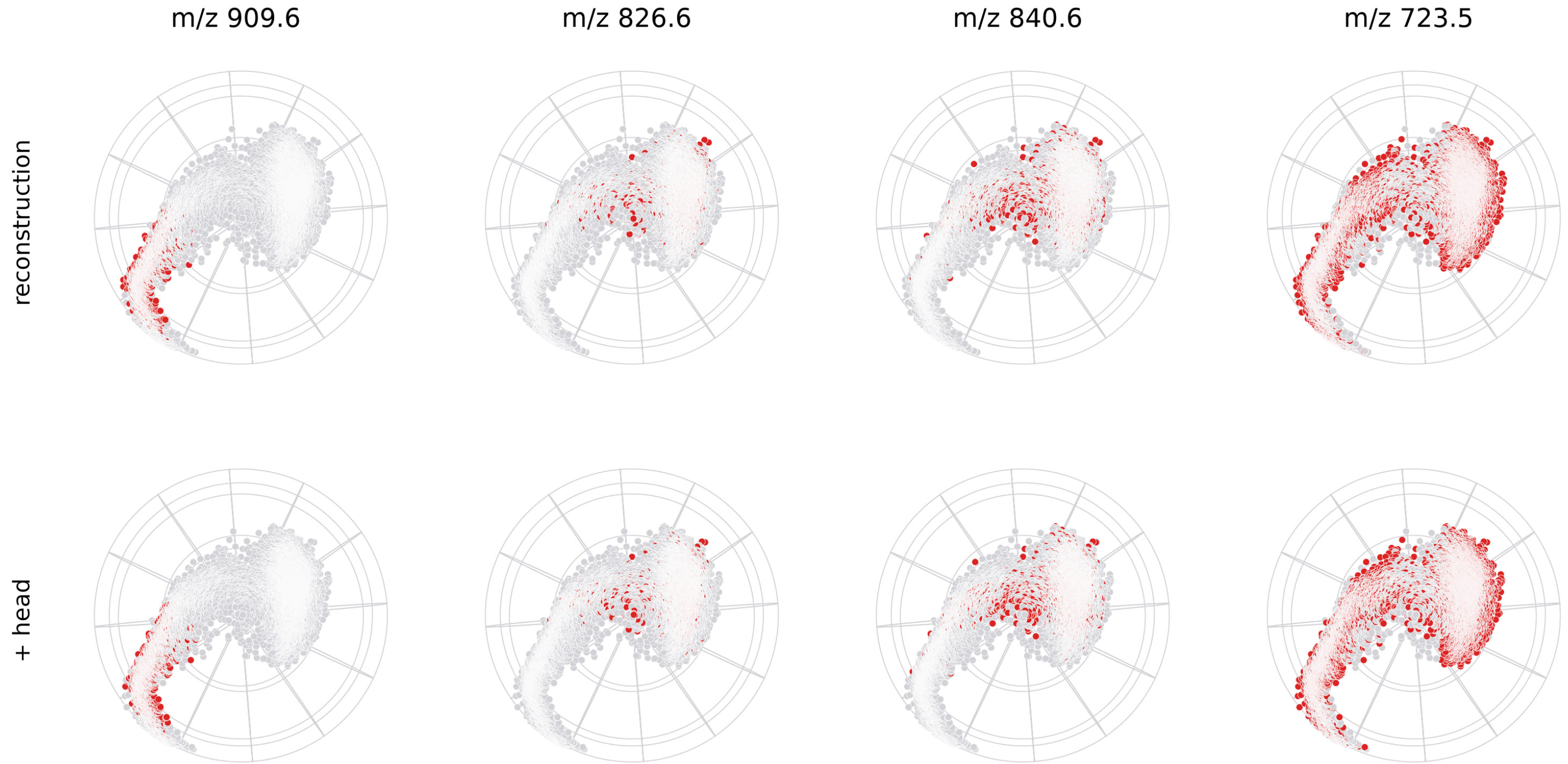


+ head



909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

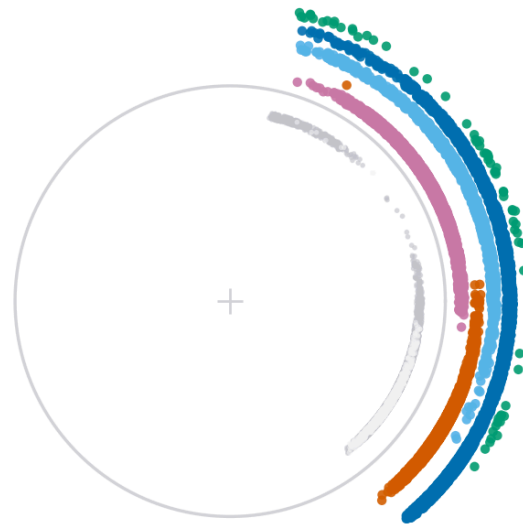
Predictive head: where the ions are



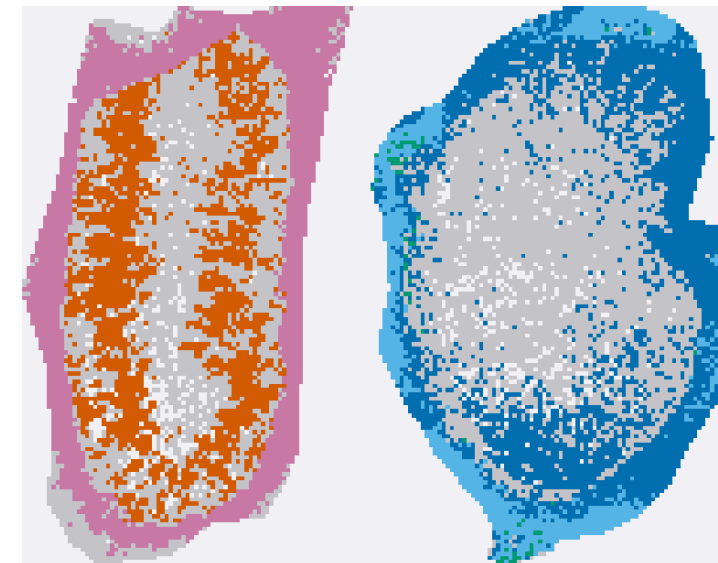
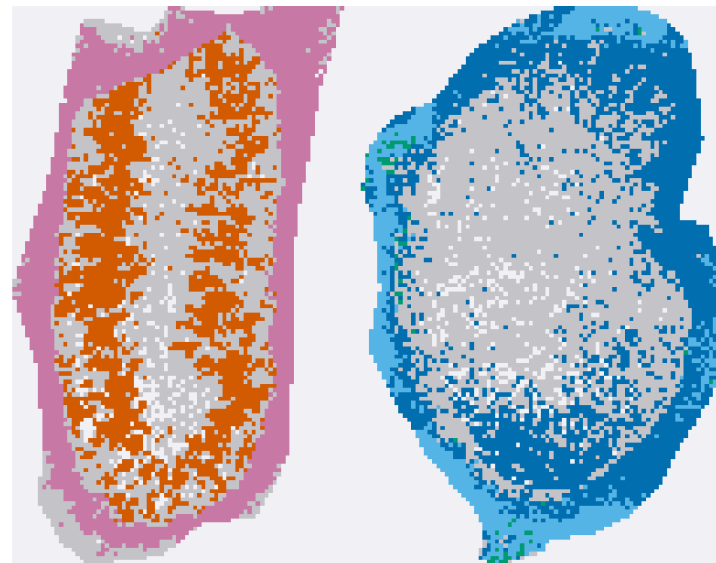
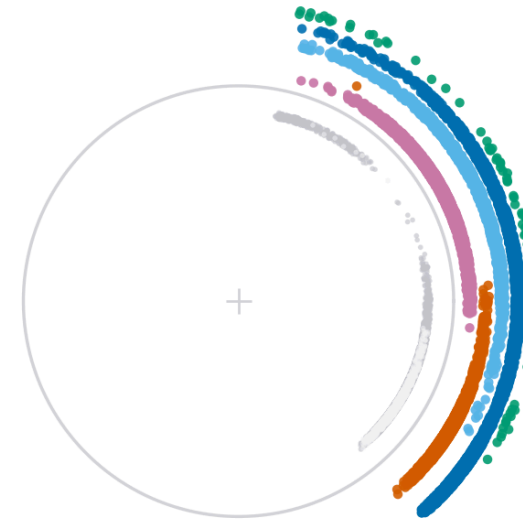
$$\mathcal{L}_{\text{VPU}} = \frac{1}{C} \sum_c \left[\log \frac{1}{|P_{\text{UU}}|} \sum_i \sigma(h_c(z_i)) - \frac{1}{|P_c|} \sum_{i \in P_c} \log \sigma(h_c(z_i)) \right] - \text{positive-unlabelled: an unannotated ion is } \textit{unknown}, \text{ not absent}$$

Contrastive learning — $\mathcal{S}^1$

+ head



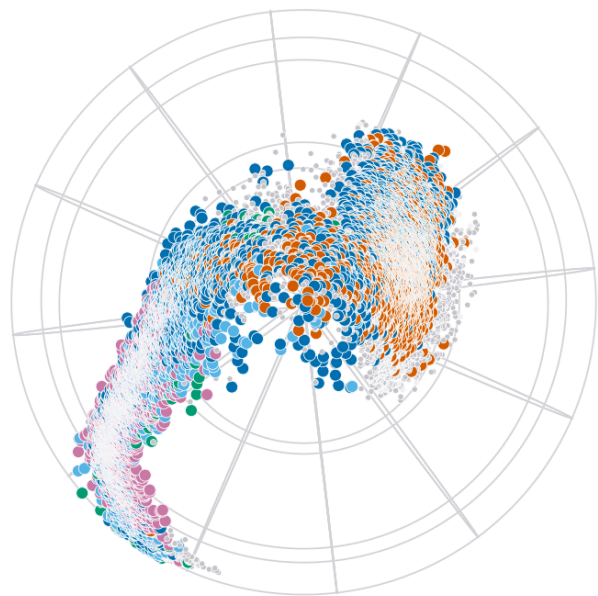
+ head + contrastive



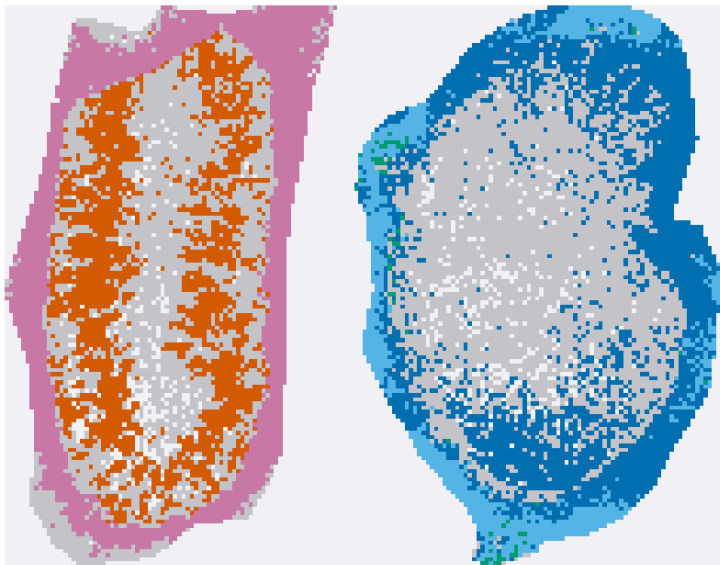
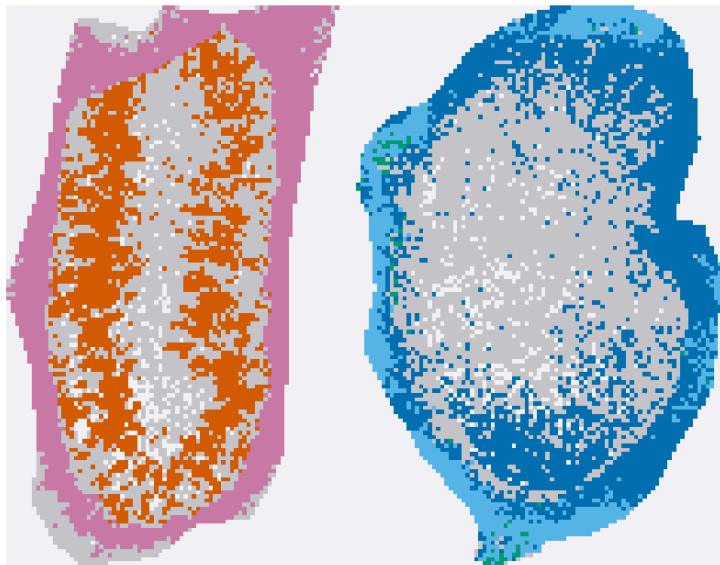
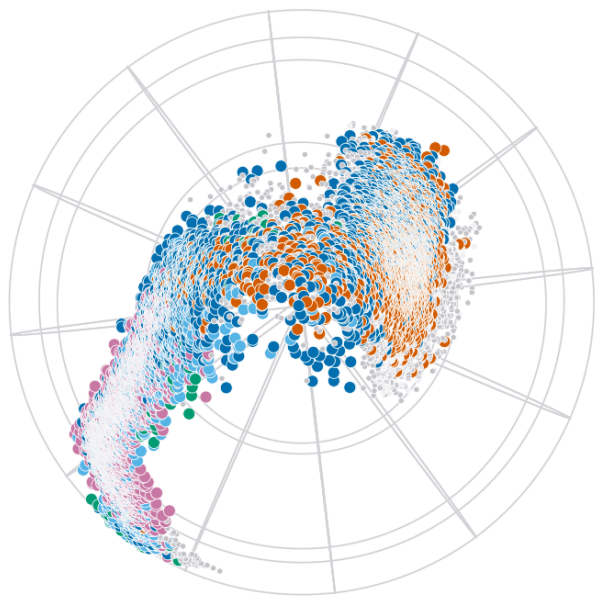
909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

Contrastive learning — $\mathcal{S}^2$

+ head

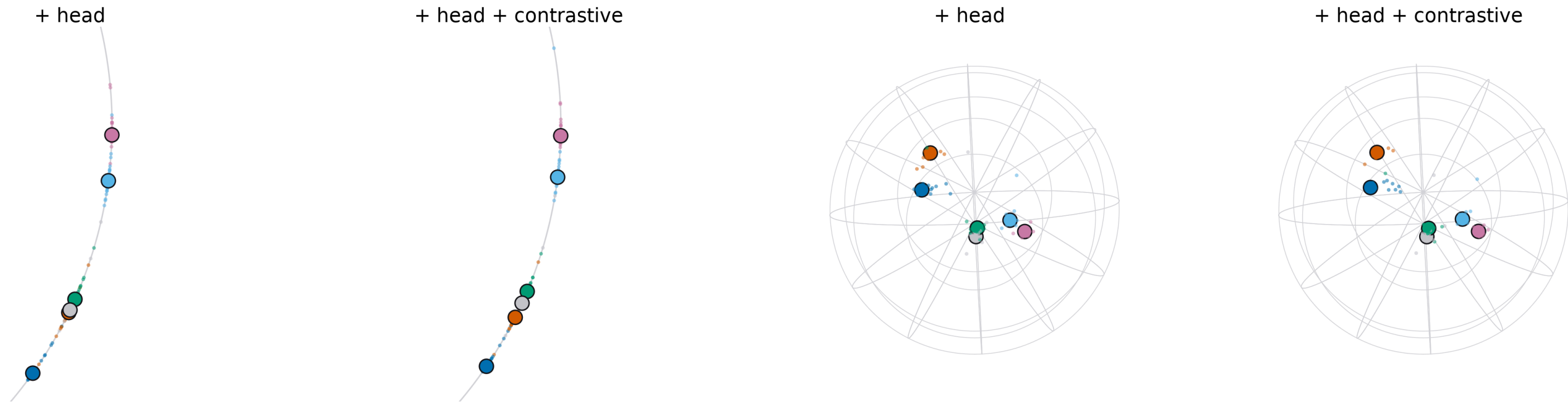


+ head + contrastive



909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

Contrastive learning: invariance to peak permutations



- a **view** $\tilde{x}$ permutes the shapes of three unannotated peak envelopes of x (TIC preserved)
- large points: pixels; small points: their views — the tighter the cloud, the more invariant the encoder

Contrastive learning: formally

FORMAL

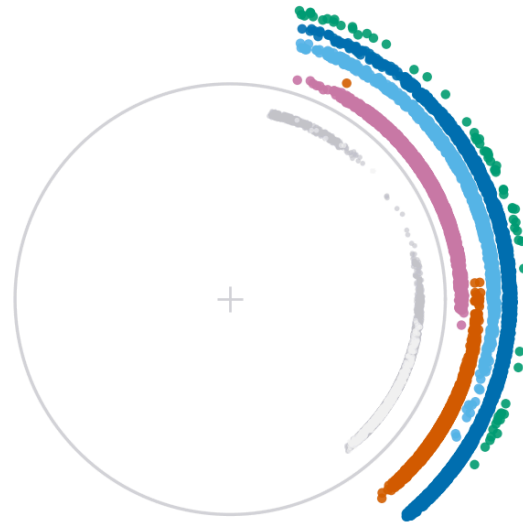
For a batch $x_1, \dots, x_B$ and views $\tilde{x}_i = T(x_i)$, projections $g_i = g(\text{Enc}(x_i))$, $g_{B+i} = g(\text{Enc}(\tilde{x}_i))$:

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2B} \sum_{i=1}^{2B} \log \frac{\exp(\text{sim}(g_i, g_{\pi(i)})/\tau)}{\sum_{j \neq i} \exp(\text{sim}(g_i, g_j)/\tau)}, \quad \text{sim}(a, b) = \frac{a^\top b}{\|a\| \|b\|}, \quad \tau = 0.07$$

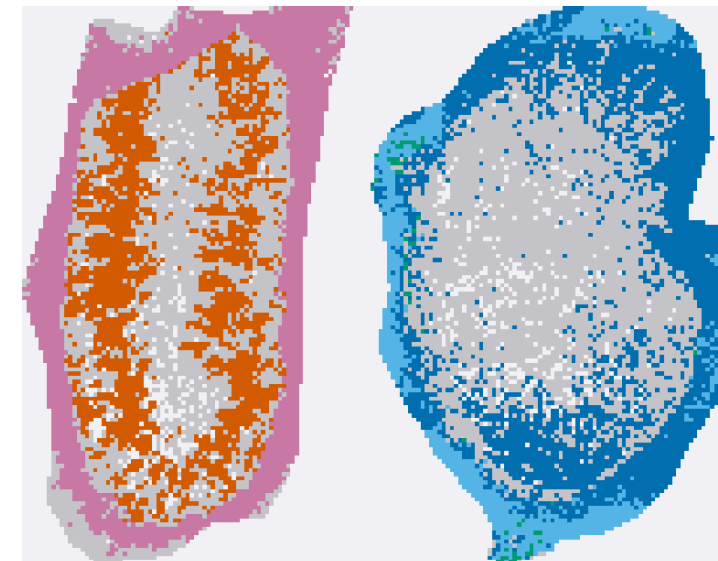
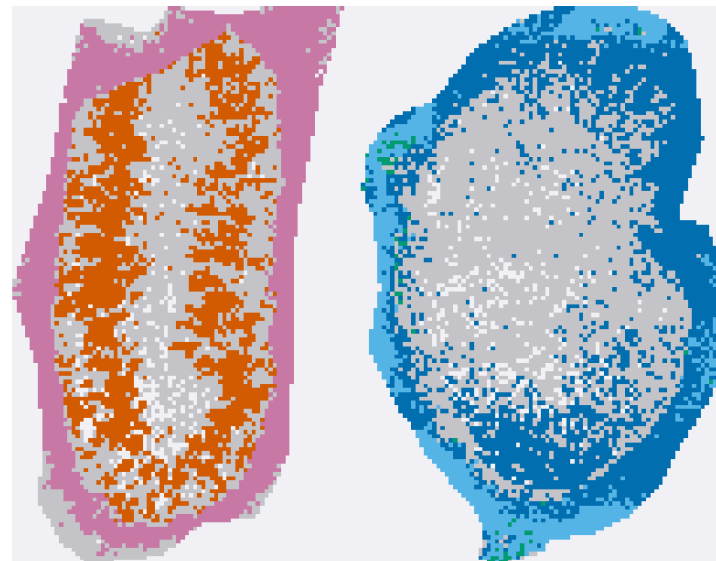
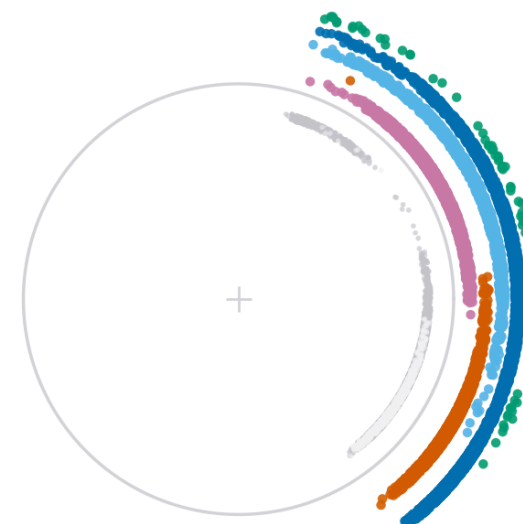
- $\pi(i)$: the other view of the same spectrum (positive pair); every other spectrum is a negative
- numerator **pulls** the two views together, denominator **pushes** different spectra apart
- T : cyclic permutation of peak-envelope shapes, **only peaks without annotation** — the representation may ignore unannotated rearrangements, but not the annotated evidence
- the loss acts on the projection $g(z)$, the latent z is shaped only through the gradient

Contractive learning — $\mathcal{S}^1$

+ head



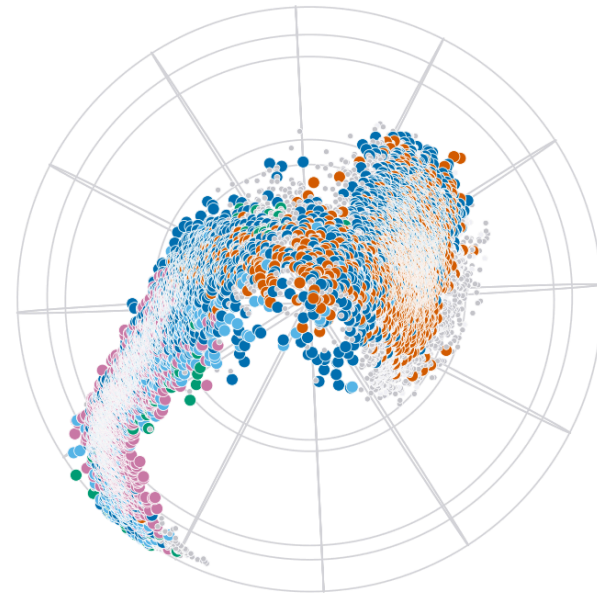
+ head + contractive



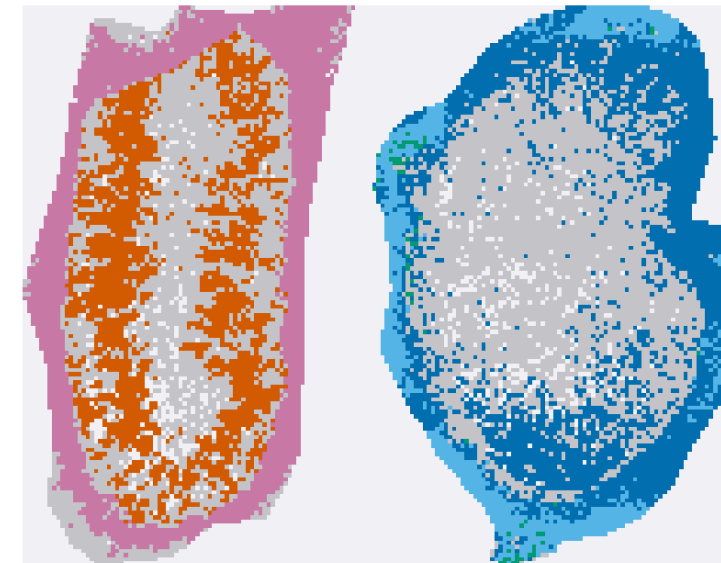
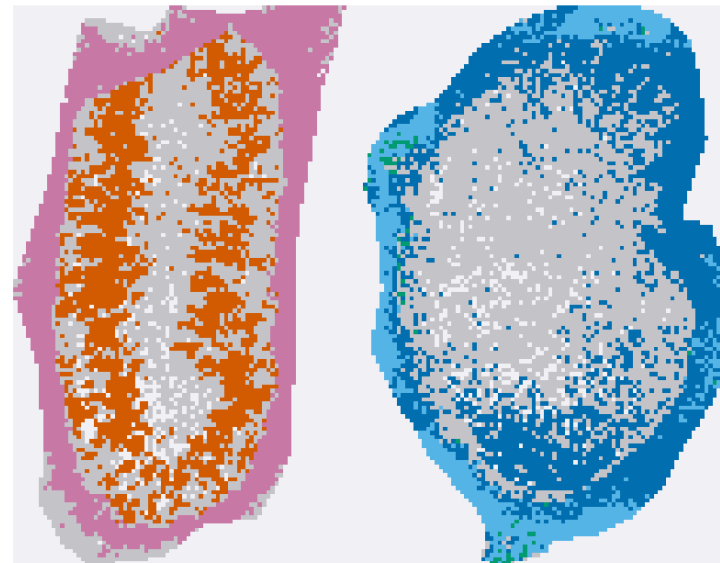
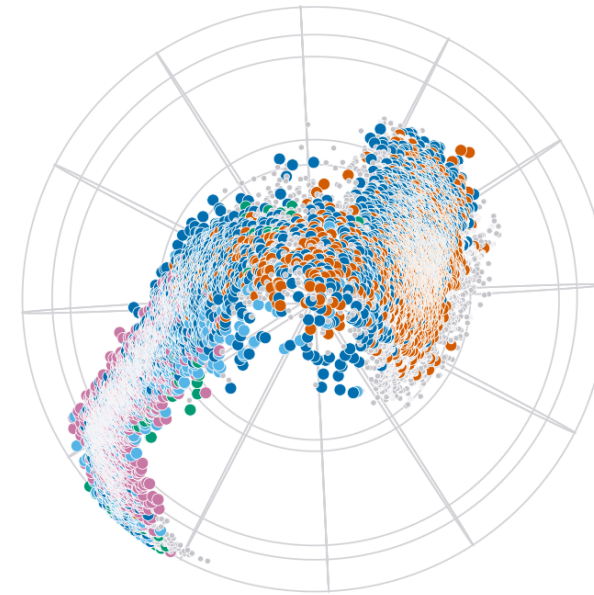
909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

Contractive learning — $\mathcal{S}^2$

+ head

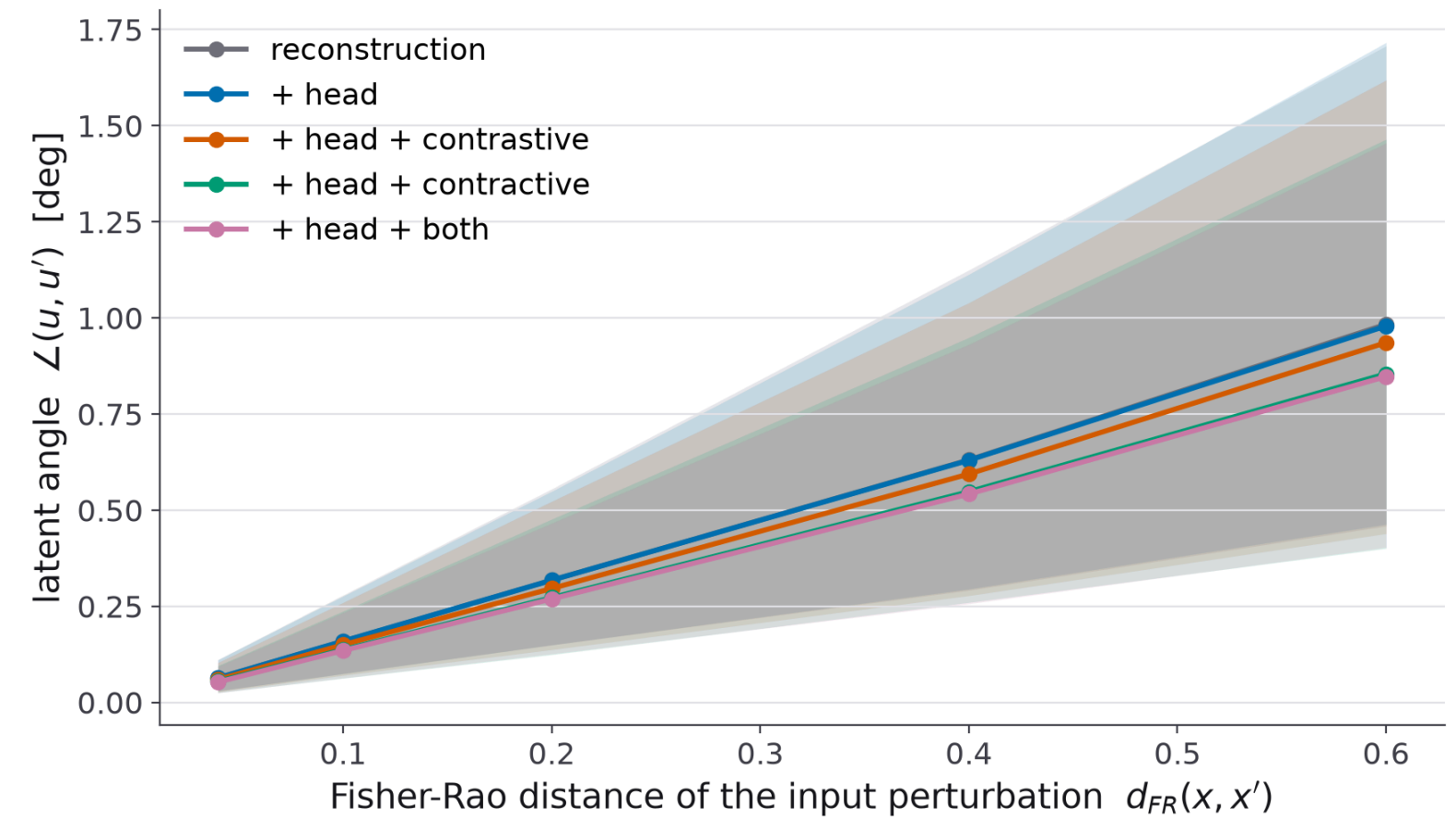
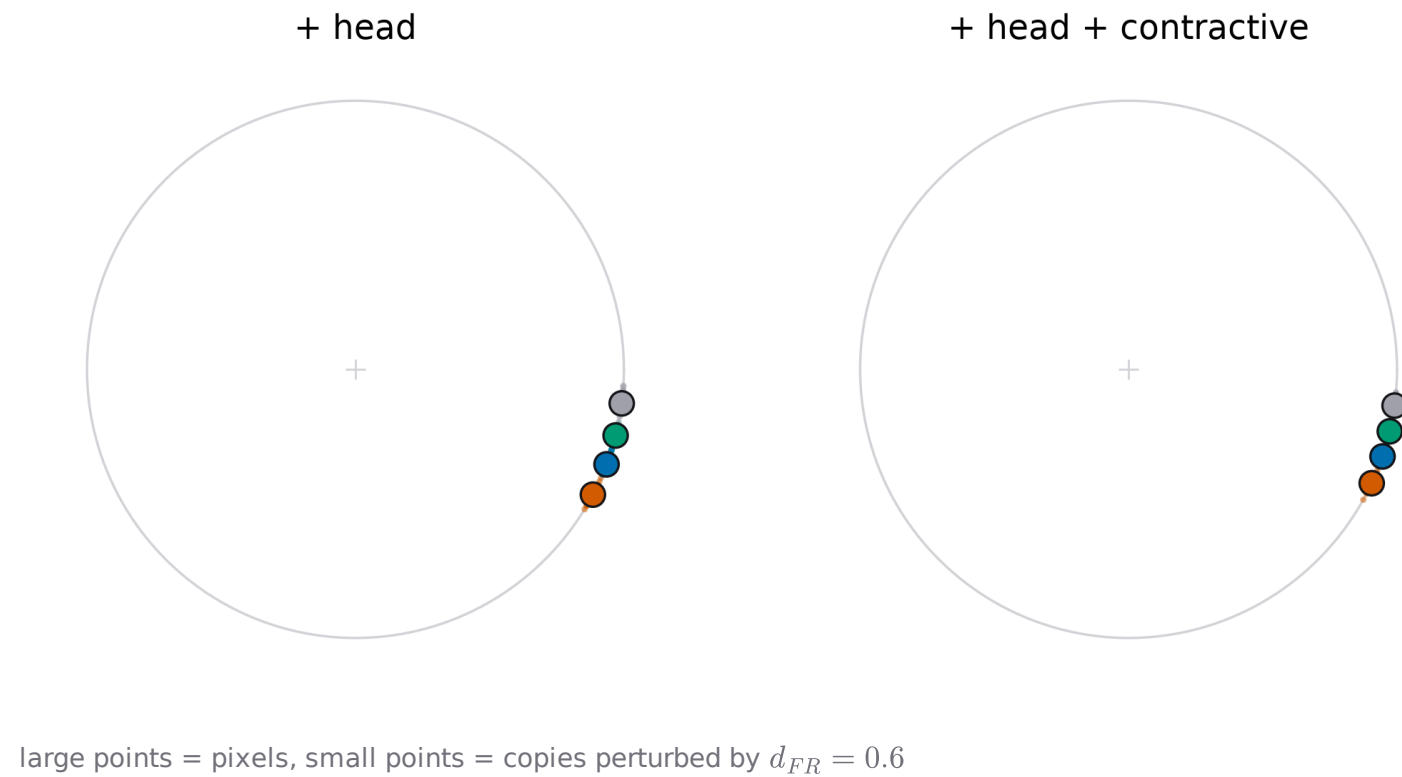


+ head + contractive



909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

Contractive learning: sensitivity — $\mathcal{S}^1$

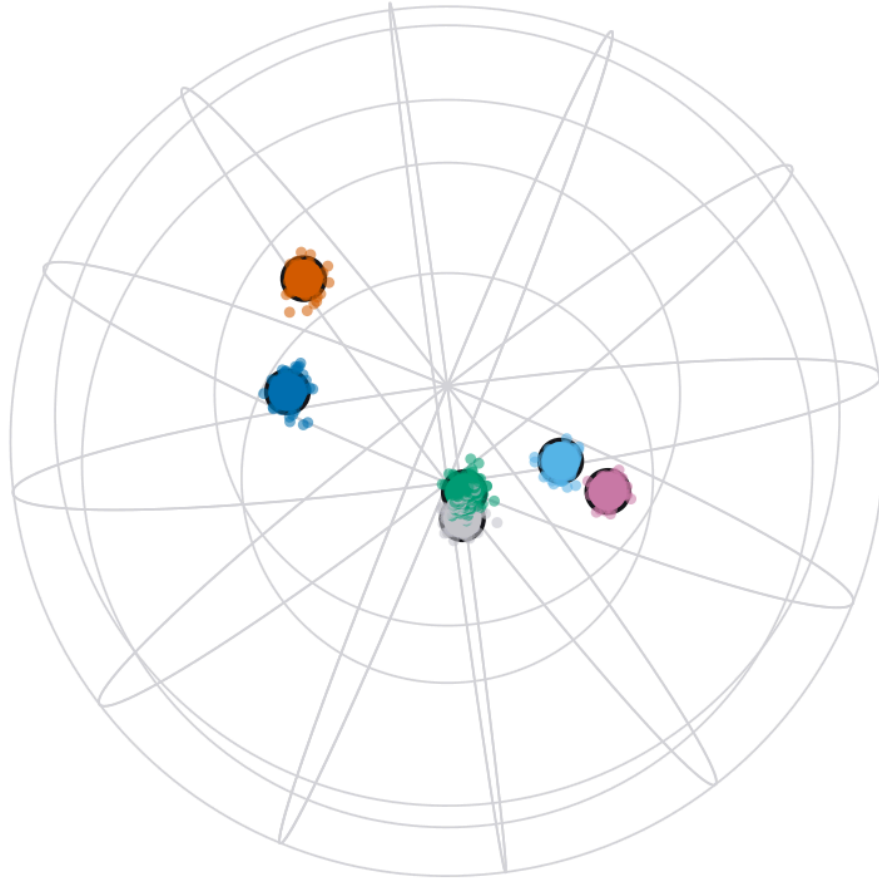


median rotation of the latent point (band: interquartile range) against the size of the input perturbation

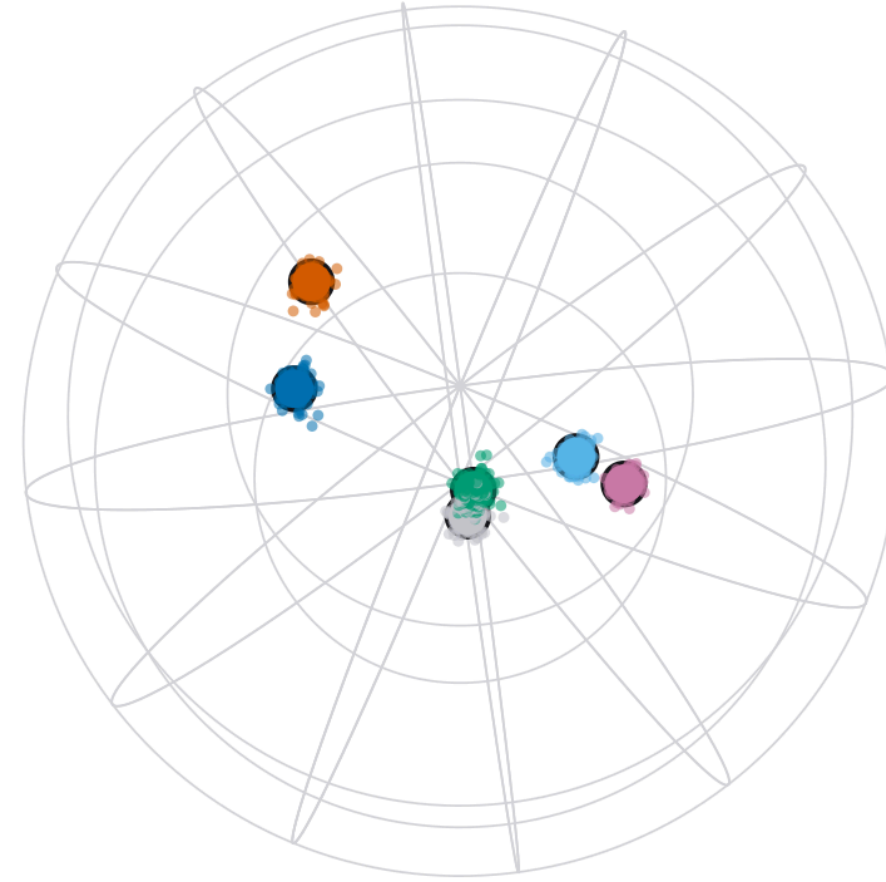
- every input spectrum is moved by a fixed Fisher-Rao distance in random directions
- the rotation of its latent point measures the local sensitivity — lower = more stable

Contractive learning: sensitivity — $\mathcal{S}^2$

+ head



+ head + contractive



- large points: pixels; small points: copies perturbed by the same Fisher-Rao distance
- the tighter the cloud, the more stable the encoder

Contractive learning: formally

FORMAL

Local sensitivity of the canonical latent: $J_u(x) = \partial u / \partial x \in \mathbb{R}^{L \times M}$

With $z = \gamma \odot u + \beta$:

$$J_z = \text{diag}(\gamma) \frac{1}{\sigma_a} P A_x, \quad P = I - \frac{1}{L} \mathbf{1} \mathbf{1}^\top - \frac{1}{L} u u^\top$$

$\|J_z\|$ can be reduced by shrinking γ without changing the encoder $\Rightarrow$ we penalize J_u .

Geometry of the input: spectra are points of the simplex; its natural metric is Fisher-Rao,

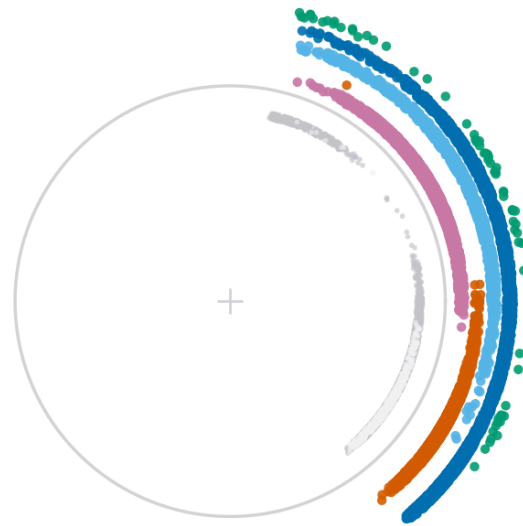
$$\langle v, w \rangle_x = \sum_i \frac{v_i w_i}{x_i}, \quad d_{FR}(x, x') = 2 \arccos \sum_i \sqrt{x_i x'_i}$$

$$A_x = J_u(x) \text{diag}(\sqrt{x}) \left(I - \sqrt{x} \sqrt{x}^\top \right), \quad \mathcal{L}_{\text{contr}} = \sigma_{\max}(A_x)^2 + \alpha \left[\sigma_{\max}(A_x) - \tau \right]_+^2$$

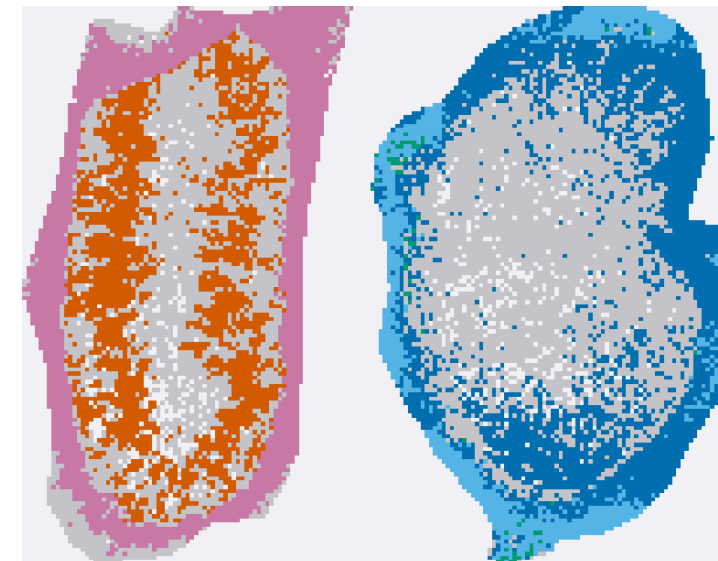
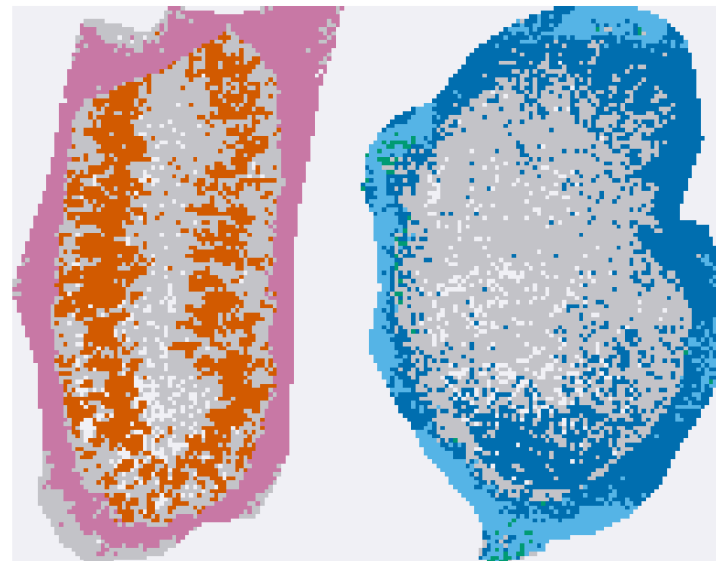
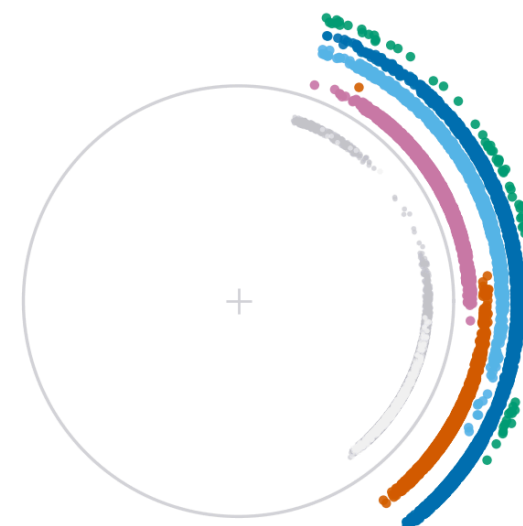
$\sigma_{\max}$ by power iteration; $\alpha = 2$, $\tau = 7$. Rifai et al. (2011) ICML — Frobenius, Euclidean, unnormalized latent.

Both regularizers — $\mathcal{S}^1$

+ head



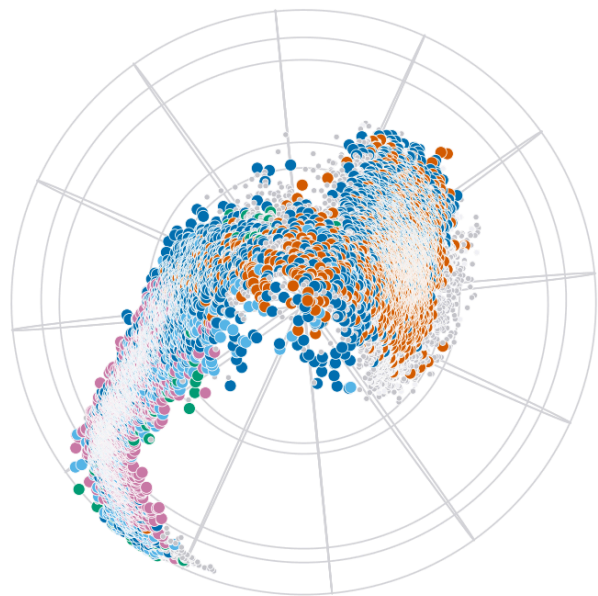
+ head + both



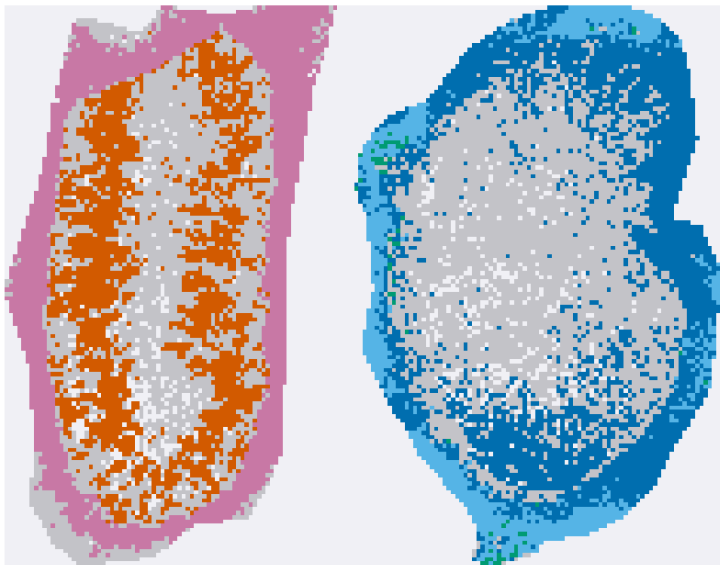
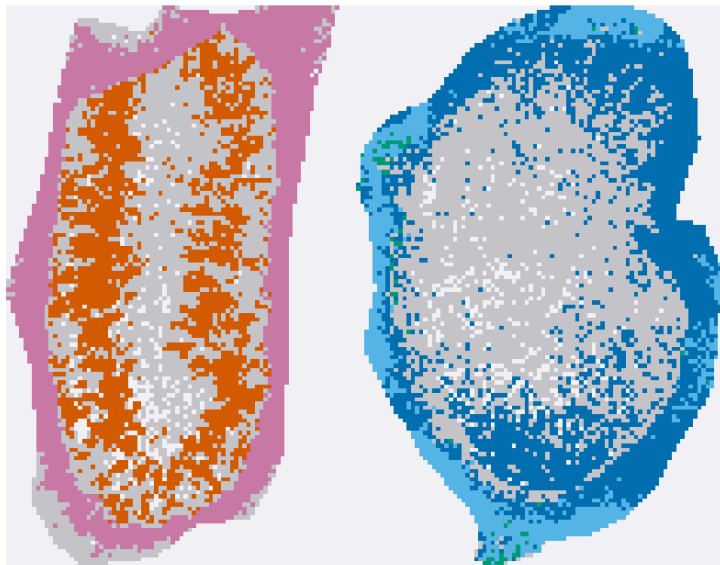
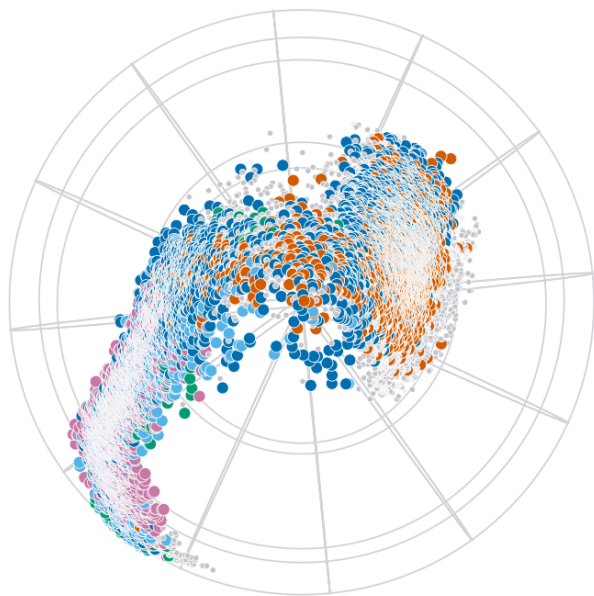
909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

Both regularizers — $\mathcal{S}^2$

+ head



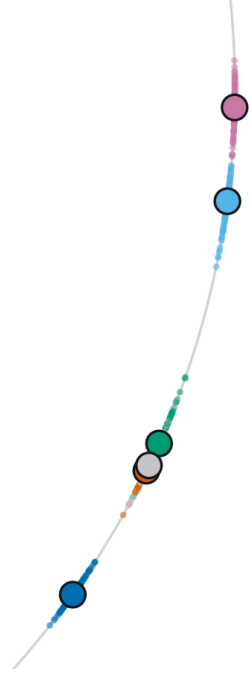
+ head + both



909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

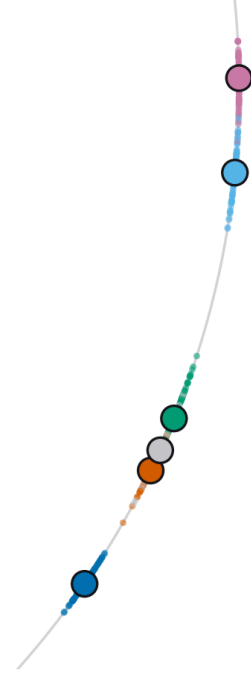
Both regularizers: stability and invariance — $\mathcal{S}^1$

+ head

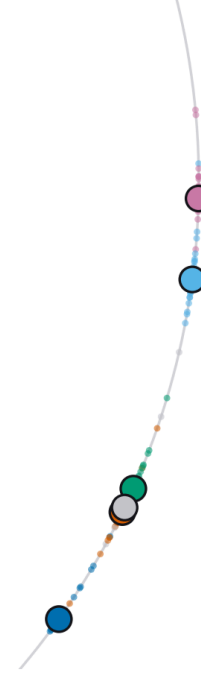


input perturbed by $d_{FR} = 0.6$ (contractive view)

+ head + both

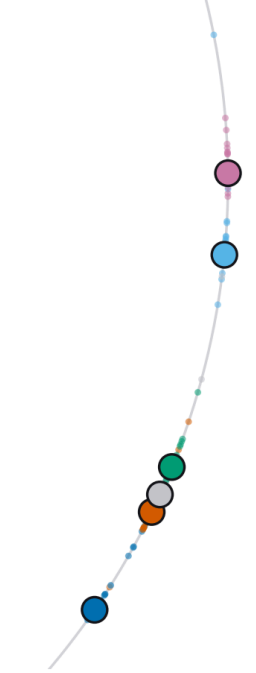


+ head



unannotated peaks permuted (contrastive view)

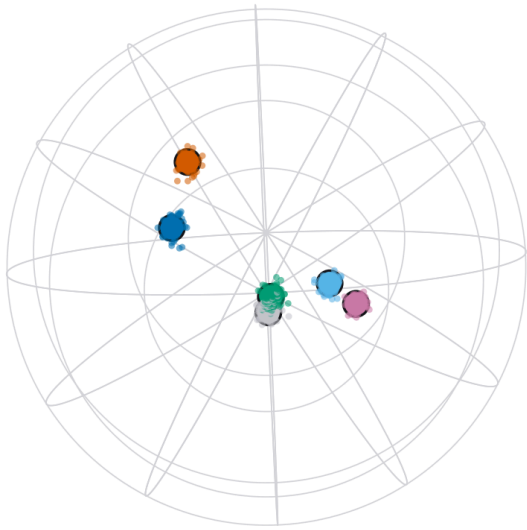
+ head + both



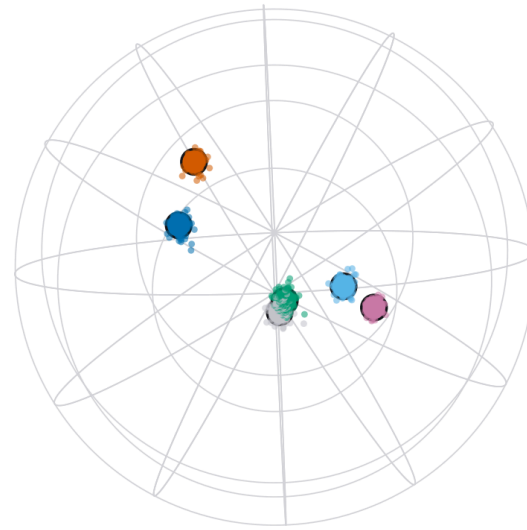
- large points: pixels; small points: perturbed copies (left) or permuted views (right)
- the tighter the cloud, the more stable (left) or invariant (right) the encoder

Both regularizers: stability and invariance — $\mathcal{S}^2$

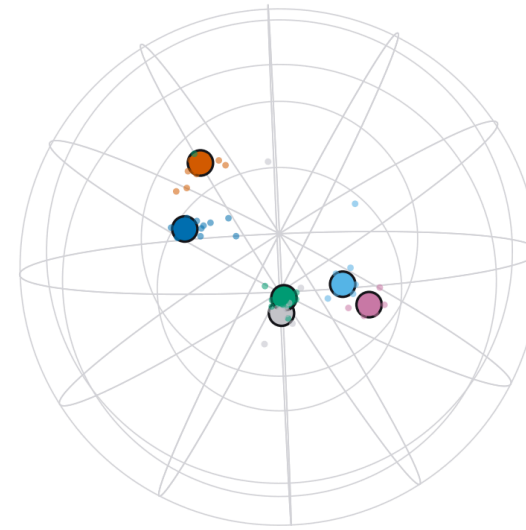
+ head



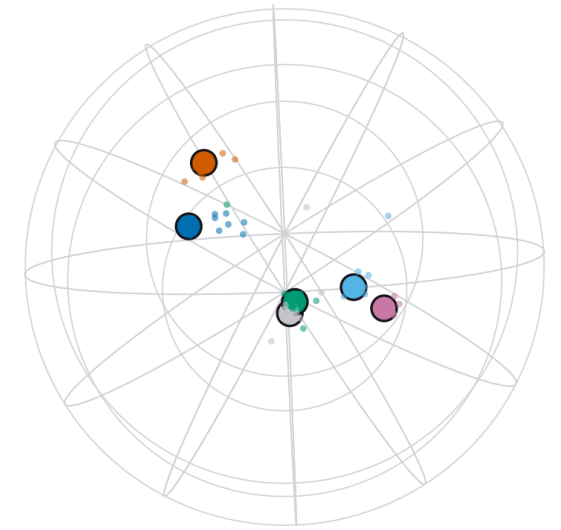
+ head + both



+ head



+ head + both



input perturbed by $d_{FR} = 0.6$ (contractive view)

unannotated peaks permuted (contrastive view)

- large points: pixels; small points: perturbed copies (left) or permuted views (right)
- the tighter the cloud, the more stable (left) or invariant (right) the encoder

Current state

Current state of the work

- representation model with predictive heads (variational PU) — prepared
- contrastive and contractive regularization — implemented and analysed
- synthetic pretraining — in progress
- next: dataset request and analysis on the benchmark dataset used by msiPL

Masserstein: comparing spectra

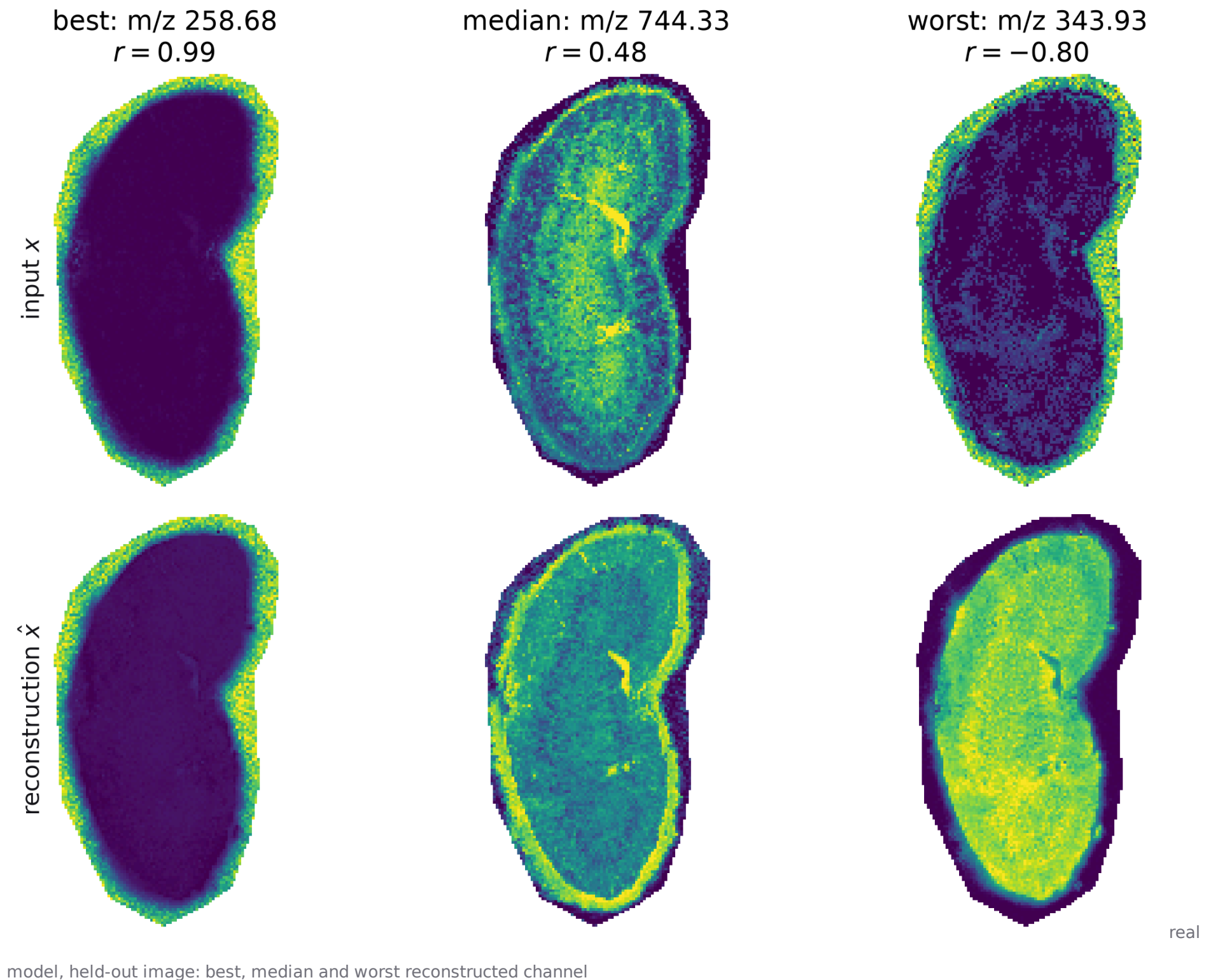
TIC-normalized spectra are probability measures on the m/z axis. We compare them by the **Wasserstein distance** — the cost of moving intensity along m/z .

In one dimension it is exact and linear-time:

$$W_1(x, \hat{x}) = \int |F_x(m) - F_{\hat{x}}(m)| dm$$

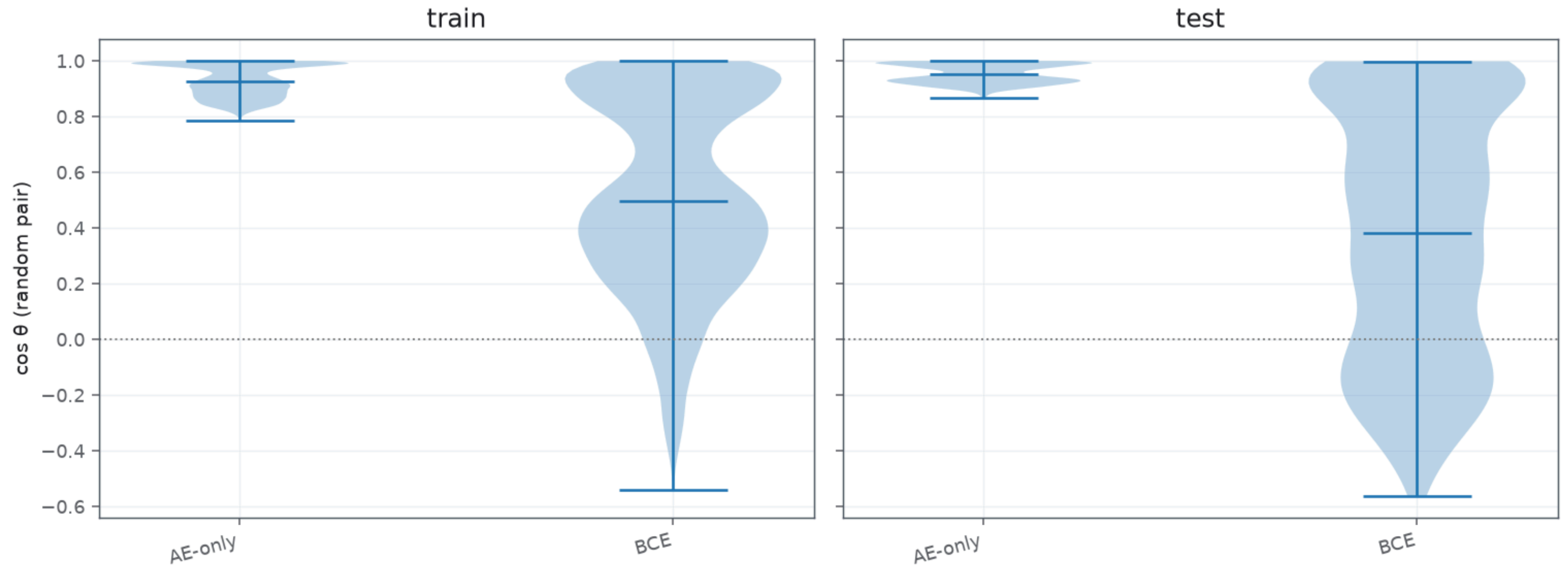
$$= \sum_{i=1}^{M-1} \left| \sum_{j \leq i} (x_j - \hat{x}_j) \right| (m_{i+1} - m_i)$$

- no transport matrix, no iterative solver: $O(M)$
- a shifted peak costs its shift, not a full mismatch



How much of the latent sphere is occupied?

Random-pair $\cos \theta$ distribution — full spread, not just mean/sd



Random-pair $\cos \theta$ distributions for train and test. Values near 1 mean two latent directions are close; a broader spread toward 0 or below indicates wider angular coverage of the sphere.

Acknowledgements

- **IsoSpec2** — Łącki, Valkenborg, Startek (2020) *Anal. Chem.*, doi:10.1021/acs.analchem.0c00959
- **Masserstein** and the one-dimensional Wasserstein distance for spectra — Majewski, Ciach, Startek et al. (2018); Ciach et al. (2020)
- the organizers of BioGeMT for the opportunity to take part